TWO SPECIAL FAMILIES OF RSK AS A LINEAR OPERATOR

BOHAO ZHANG

Abstract. We study the properties of two families of linear operators from the Robinson–Schensted–Knuth correspondence with fixed row and column sums.The Robinson–Schensted–Knuth correspondence, also known as the RSK algorithm, is a crucial combinatorial model which builds a bijection from the matrix with non negative integer entries to a pair of semi-standard Young Tableaux of the same shape. For the first family, $A_n = \mathrm{RSK}_{(2n-2,2),(2^n)}$, we obtain an explicit matrix description and identify its action with the symmetric square of an operator similar to an n-cycle permutation matrix. Consequently, A_n is diagonalizable over $\mathbb{C}$ and all its eigenvalues are n-th roots of unity. For odd $n > 3$, we prove that$\chi_{A_n}(t) = (t^n - 1)^{(n+1)/2}$. For the second family, $\mathrm{RSK}_{(1,a,a),(a,a,1)}$, we investigate an explicit polynomial $F_a(t)$ of degree $2a + 2$. Computations for $15 \le a \le 100$ indicate that $F_a(t)$ divides the corresponding characteristic polynomial and has Galois group S_{2a+2} over $\mathbb{Q}$. We also have shown an approach to its irreducibility using Rouché's theorem and estimates on a circle centered at -2.

1. Introduction

This report studies two families of RSK linear operators with fixed row and column sums, focusing on their eigenvalues and characteristic polynomials. The Robinson–Schensted–Knuth (RSK) correspondence is a bijection between matrices with non-negative integers and pairs of semistandard Young tableaux of the same shape. It was used to as a powerful and easy way to prove the Cauchy Identity[Sta01], which is of importance. It is also well studied for several crucial variants, such as skew RSK[SS90], dual RSK[But94]. Its linear realization in [SY26] raises questions about diagonalizability and the arithmetic properties of eigenvalues. For the first family, we obtain a uniform description in terms of roots of unity. For the second family, we prove the irreducibility of a polynomial family suggested by computations of characteristic polynomials.

Our first family is

$$A_n = \mathrm{RSK}_{(2n-2,2),(2^n)}, \qquad n > 3,$$

where (2^n) denotes the composition with n entries equal to 2. We identify A_n as the one of the images of a representation of symmetric space of a linear operator on $\mathbb{C}^n$ that is similar to an n-cycle permutation matrix. This construction of the representation gives the following result.

Theorem 1.1. *For every integer $n > 3$, the operator A_n is diagonalizable over $\mathbb{C}$ and all its eigenvalues are n-th roots of unity. If n is odd, then*

$$\chi_{A_n}(t) = (t^n - 1)^{(n+1)/2}.$$

In particular, each n-th root of unity occurs with multiplicity $(n + 1)/2$.

The second family,

$$T_a = \mathrm{RSK}_{(1,a,a),(a,a,1)},$$

exhibits a different pattern in computations. Let $\chi_a(t)$ denote its characteristic polynomial, and, for an integer $a \ge 2$, define

$$F_a(t) = t(t+2)(t^2+1)(t^2-1)^{a-1} + (-1)^a \frac{t^4 + 2t^3 - t + 1 - t^{2a+2}(t+2)}{t^2 - 1}.$$

Date: Sepetember 15, 2026 ICLUE 2026 Progress Report.

Computations for $15 \le a \le 100$ indicate that

$$F_a(t) \mid \chi_a(t), \qquad \mathrm{Gal}(E_a/\mathbb{Q}) \cong S_{2a+2},$$

where E_a is the splitting field of $F_a(t)$ over $\mathbb{Q}$. These observations suggest the most of eigenvalues that cannot be expressed by radicals and motivate the study of $F_a(t)$ beyond the computed range. Our second main result shows its irreducibility for all sufficiently large a.

Theorem 1.2. *For every integer $a \ge 100$, the polynomial $F_a(t)$ is irreducible over $\mathbb{Q}$.*

The proof is inspired by the approach of Weilin Zhang and Pingzhi Yuan in [WP23], which combines information about the locations of polynomial roots with arithmetic restrictions on possible factors. In our setting, Rouché's theorem shows that $F_a(t)$ has exactly one root, counted with multiplicity, in the disk $|t+2| < 1$, and none on its boundary. We then use $F_a(-2) = (-1)^a$ to rule out a nontrivial factorization in $\mathbb{Z}[t]$.

Section 2 reviews the RSK correspondence and its linear realization, then develops the matrix description and symmetric-square construction for A_n. Section 3 uses this construction to prove the characteristic-polynomial formula in Theorem 1.1. Section 4 presents the computations for the second family and proves Theorem 1.2 through estimates on the roots of $F_a(t)$.

1.1. **Outline.** Section 2 reviews Young tableaux, the RSK correspondence, and its realization as a linear operator through the monomial and standard bitableau bases. It then develops an block description of $A_n = \mathrm{RSK}_{(2n-2,2),(2^n)}$ and identifies A_n with the symmetric square of an operator similar to a cyclic permutation. Section 3 uses this discussion to determine the characteristic polynomial of A_n for odd $n > 3$. Finally, Section 4 studies the family $\mathrm{RSK}_{(1,a,a),(a,a,1)}$. It presents computational evidence concerning a factor $F_a(t)$ of its characteristic polynomial and the Galois group of this factor, and investigates an approach to irreducibility based on Rouché's theorem.

Acknowledgments

The author thanks Alexander Yong for guidance and helpful discussions, and ICLUE2026 for its support. ?Add funding information only if applicable?.

2. Preliminary Notions and Facts

2.1. **Basic definitions.** We shall recall some basic definitions of some combinatorical models.

Definition 2.1.1. Let λ be a partition. A *Young tableau* of shape λ is a filling of the Young diagram of λ with positive integers. A *semistandard Young tableau* (SSYT) is a Young tableau whose entries weakly increase from left to right, also each row and strictly increase from up to down along each column. A *standard Young tableau* (SYT) is an SSYT in which $1, \ldots, |\lambda|$ are all filled in.

Example 2.1.2. The following are tableaux of shape $\lambda = (3, 2)$:

3	1	2
2	1	

Young tableau

1	1	3
2	3	

Semistandard Young tableau

1	2	4
3	5	

Standard Young tableau

The first tableau is not semistandard. The second is semistandard because its rows are weakly increasing and its columns are strictly increasing, but it is not standard because some entries repeat. The third is standard because it is semistandard and contains each of $1, \ldots, 5$ exactly once.

2.2. **The Robinson–Schensted–Knuth correspondence.** The Robinson–Schensted–Knuth (RSK) correspondence is a bijection

$$A = (a_{ij}) \in \mathbb{Z}_{\geq 0}^{m \times n} \quad \longleftrightarrow \quad (P, Q),$$

where P and Q are semistandard Young tableaux of the same shape, with entries in $\{1, \ldots, n\}$ and $\{1, \ldots, m\}$, respectively.

Forward direction: from a matrix to tableaux. Given $A = (a_{ij})$, form a biword

$$\begin{pmatrix} i_1 & i_2 & \cdots & i_N \\ j_1 & j_2 & \cdots & j_N \end{pmatrix}, \qquad N = \sum_{i,j} a_{ij},$$

in which each pair (i, j) occurs a_{ij} times, starting from left to right, up to down.

We now associate with A a pair (P, Q) of SSYTs of the same shape, as follows. Let

$$w_A = \begin{pmatrix} i_1 & i_2 & \cdots & i_m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix}.$$

Begin with $(P(0), Q(0)) = (\varnothing, \varnothing)$ (where $\varnothing$ denotes the empty SSYT). If $t < m$ and $(P(t), Q(t))$ are defined, then let

(a) $P(t+1) = P(t) \leftarrow j_{t+1}$;
(b) $Q(t+1)$ be obtained from $Q(t)$ by inserting i_{t+1} (leaving all parts of $Q(t)$ unchanged) so that $P(t+1)$ and $Q(t+1)$ have the same shape.

The process ends at $(P(m), Q(m))$, and we define $(P, Q) = (P(m), Q(m))$. We denote this correspondence by $A \xrightarrow{\text{RSK}} (P, Q)$ and call it the *RSK algorithm*. We call P the *insertion tableau* and Q the *recording tableau* of A or of w_A.

Reverse direction: from tableaux to a matrix. Given semistandard tableaux (P, Q) of the same shape, start with the zero matrix A and repeat the following until both tableaux are empty:

(1) Let i be the largest entry of Q. Among the boxes containing i, choose the rightmost one. This box is an outer corner. Remove it from Q.
(2) Remove the corresponding box from P, and let x be its entry. Perform reverse row insertion: in the row immediately above, replace the rightmost entry $y < x$ by x, then set $x = y$. Continue upward in this way. be the entry carried out of that row. If the removed box was already in the first row, let j be its entry. Increase a_{ij} by 1.

The recovered pairs (i, j) occur in the reverse order of the original biword, and the resulting matrix is the inverse RSK image of (P, Q).

2.3. **The RSK correspondence as a linear operator.** For interested readers, one can refer to [SY26] for more details.

Let $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_m)$ and $\beta = (\beta_1, \beta_2, \ldots, \beta_n)$ be two given compositions of the same integer N:

$$|\alpha| = \sum_{i=1}^{m} \alpha_i = \sum_{j=1}^{n} \beta_j = |\beta| = N$$

We define the set of non-negative integer matrices with row sums restricted by α and column sums restricted by β as:

$$\mathcal{M}(\alpha, \beta) = \left\{ A = (a_{ij}) \in \mathrm{M}_{m \times n}(\mathbb{N}_0) \;\middle|\; \sum_{j=1}^{n} a_{ij} = \alpha_i, \; \sum_{i=1}^{m} a_{ij} = \beta_j \right\}$$

Since the matrix entries $a_{ij} \in \mathbb{N}_0$ are bounded by the fixed row and column sums, satisfying $0 \leq a_{ij} \leq \min(\alpha_i, \beta_j)$, the cardinality of this set is finite.

The complex vector space generated by this finite set as a basis can be expressed as:

$$V_{(\alpha,\beta)} = \mathbb{C}[\mathcal{M}(\alpha,\beta)] = \bigoplus_{A \in \mathcal{M}(\alpha,\beta)} \mathbb{C} \cdot A$$

This space $V_{(\alpha,\beta)}$ is a finite-dimensional $\mathbb{C}$-vector space generated by all non-negative integer matrices with the given fixed row and column sums. Its dimension is:

$$\dim_{\mathbb{C}} V_{(\alpha,\beta)} = |\mathcal{M}(\alpha,\beta)|$$

.

Remark 2.3.1. In the following of the report , we shall abuse the notation of z^A and A, for example, we view $z^{\begin{pmatrix}1&1&0\\1&0&2\end{pmatrix}} = z_{11}z_{12}z_{21}z_{23}^2 = \begin{pmatrix}1&1&0\\1&0&2\end{pmatrix} \in V_{(\alpha,\beta)}$

Now for a given pair of compostions α and β such that $|\alpha| = |\beta| = N$, we take a matrix $A \in \mathcal{M}(\alpha,\beta)$, and we do the shown RSK algorithm on A to get We recall the bitableau basis referenced in the introduction, now in the context of our space of $m \times n$ matrices. Let $\Delta_1, \ldots, \Delta_N$ be a sequence of minors of the generic $m \times n$ matrix

$$Z = [z_{ij}]_{1 \le i \le m, 1 \le j \le n}.$$

We may assume that the respective sizes of the minors are weakly decreasing. We can encode the product

$$\Delta_1 \ldots \Delta_N$$

as a pair of semistandard Young tableaux (P, Q), where the c-th columns (from the left) of P and Q are filled by the row and column indices of Δ_c respectively.

When applying the RSK correspondence to a matrix $A \in \mathcal{M}(\alpha,\beta)$, the resulting tableaux P and Q are guaranteed to be semistandard with weights corresponding to the compositions β and α. When P and Q are both semistandard, we call the corresponding product of minors a *standard bitableau* and denote it $[P|Q]$.

Lemma 2.3.2. *([SY26], Lemma 2.5) The standard bitableaux $[P|Q]$ with cotent (σ,π) form a linear basis of $R_(\sigma,\pi)$*

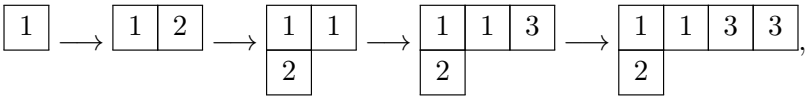
Example 2.3.3. Consider

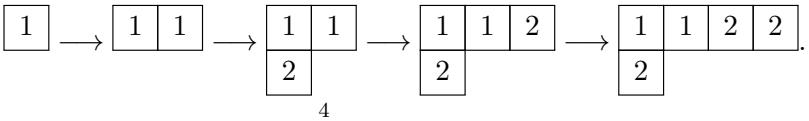
$$A = \begin{pmatrix}1&1&0\\1&0&2\end{pmatrix}.$$

Its biword is

$$\begin{pmatrix}1&1&2&2&2\\1&2&1&3&3\end{pmatrix}.$$

Thus we insert the word $1,2,1,3$ into P and record the entries $1,1,2,2$ in Q:

P : $\varnothing \longrightarrow$  ,

Q : $\varnothing \longrightarrow$  .

The inverse procedure is shown below. Each arrow is labelled by the recovered pair (i, j):

$$\left(\begin{array}{|c|c|c|c|}\hline 1&1&3&3\\\hline 2\\\cline{1-1}\end{array},\ \begin{array}{|c|c|c|c|}\hline 1&1&2&2\\\hline 2\\\cline{1-1}\end{array}\right) \xrightarrow{(2,3)} \left(\begin{array}{|c|c|c|}\hline 1&1&3\\\hline 2\\\cline{1-1}\end{array},\ \begin{array}{|c|c|c|}\hline 1&1&2\\\hline 2\\\cline{1-1}\end{array}\right) \xrightarrow{(2,3)} \left(\begin{array}{|c|c|}\hline 1&1\\\hline 2\\\cline{1-1}\end{array},\ \begin{array}{|c|c|}\hline 1&1\\\hline 2\\\cline{1-1}\end{array}\right)$$

$$\xrightarrow{(2,1)} \left(\begin{array}{|c|c|}\hline 1&2\\\hline\end{array},\ \begin{array}{|c|c|}\hline 1&1\\\hline\end{array}\right)$$

$$\xrightarrow{(1,2)} \left(\begin{array}{|c|}\hline 1\\\hline\end{array},\ \begin{array}{|c|}\hline 1\\\hline\end{array}\right) \xrightarrow{(1,1)} (\varnothing, \varnothing).$$

Counting these pairs recovers the original matrix A. Now, we write down the $[P|Q]$,

$$[P|Q] = \begin{vmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{vmatrix} z_{11}z_{32}z_{32} = z_{11}^2 z_{22} z_{32}^2 - z_{12}z_{21}z_{32}^2 \in V_{((2,3)(2,1,2))}$$

.

In [SY26], some problems about diagonalizability and the characteristic about roots about taking RSK algorithm as a linear operator when taking a pair of compositions are still open. In our report, we provide two families of RSK matrix to shot that even when taking a pair of compositions, It could happen that either be all the roots of the unity, or one can conjecture that most of the roots can not be expressed by radicals.

2.4. **A preliminary lemma.** The following lemma gives an explicit block description of the first family of RSK matrices considered in this report.

Lemma 2.4.1. *Let*

$$A_n := \mathrm{RSK}_{(2n-2,2),(2^n)}, \qquad n > 3,$$

and put

$$N = \binom{n+1}{2}.$$

Then A_n has the block form

$$A_n = \begin{pmatrix} U_n \\ L_n \end{pmatrix},$$

where $L_n \in M_{n\times N}(\mathbb{Z})$ and $U_n \in M_{\binom{n}{2}\times N}(\mathbb{Z})$.

Divide the columns into $n-1$ consecutive blocks of lengths

$$\ell_1 = 3, \qquad \ell_j = j+1 \quad (2 \le j \le n-1).$$

Thus

$$\ell_1 + \ell_2 + \cdots + \ell_{n-1} = 3 + 3 + 4 + \cdots + n = \binom{n+1}{2} = N.$$

With respect to this column decomposition, the upper block is

$$U_n = \mathrm{diag}(C_1, C_2, \ldots, C_{n-1}),$$

where $C_1 = (1,1,1)$, and, for $2 \le j \le n-1$, $C_j \in M_{j\times(j+1)}(\mathbb{Z})$ is defined by

$$(C_j)_{ab} = \begin{cases} 1, & a = 1 \text{ and } b \in \{1,2\}, \\ 1, & 2 \le a \le j \text{ and } b = a+1, \\ 0, & \text{otherwise.} \end{cases}$$

Write the lower block as

$$L_n = \left(L_n^{(1)} \mid L_n^{(2)} \mid \cdots \mid L_n^{(n-1)}\right),$$

where $L_n^{(1)} \in M_{n\times 3}(\mathbb{Z})$ *and* $L_n^{(j)} \in M_{n\times(j+1)}(\mathbb{Z})$ *for* $2 \le j \le n-1$. *For every* j, *write*

$$L_n^{(j)} = \begin{pmatrix} R_1^{(j)} \\ R_2^{(j)} \\ \vdots \\ R_{n-1}^{(j)} \\ p^{(j)} \end{pmatrix}.$$

The last row is

$$p^{(1)} = (0,0,1), \qquad p^{(j)} = (0, \underbrace{1,\dots,1}_{j \text{ entries}}) \quad (2 \le j \le n-1).$$

For the first $n-1$ *rows, set* $R_i^{(j)} = 0$ *if* $j < i$. *On the diagonal blocks,*

$$R_1^{(1)} = (0,-1,-2), \qquad R_i^{(i)} = (0, \underbrace{-1,\dots,-1}_{i-1 \text{ entries}}, -2) \quad (2 \le i \le n-1).$$

For the blocks strictly above the diagonal, set

$$R_1^{(j)} = (-1,-1,0,\dots,0) \quad (2 \le j \le n-1),$$

and

$$R_i^{(j)} = -e_{i+1}^{(j+1)} \quad (2 \le i < j \le n-1),$$

where $e_{i+1}^{(j+1)}$ *denotes the* $(i+1)$*-st standard row vector of length* $j+1$.

Proof. Let

$$\sigma = (2n-2,2), \qquad \pi = (2,2,\dots,2) = (2^n).$$

For $1 \le i \le j \le n$, define

$$m_{i,j} = z_{1,1}^2 z_{1,2}^2 \cdots z_{1,n}^2 \frac{z_{2,i} z_{2,j}}{z_{1,i} z_{1,j}}.$$

An ordered monomial basis of the vector space $V_{(\sigma,\pi)}$ is

$$\begin{aligned}\mathcal{B}_n = \big(&m_{n,n};\ m_{n-1,n}, m_{n-1,n-1};\ m_{n-2,n}, m_{n-2,n-1}, m_{n-2,n-2};\\ &\dots;\ m_{1,n}, m_{1,n-1}, \dots, m_{1,1}\big).\end{aligned}$$

Denote the linear operator $\mathrm{RSK}_{\sigma,\pi}$ by T. Applying RSK to the basis monomials gives

$$T(m_{n,n}) = m_{n,n}.$$

For $1 \le a \le n-1$,

$$T(m_{a,n}) = m_{a+1,n} - m_{1,n}.$$

For $1 \le a < b \le n-1$,

$$T(m_{a,b}) = m_{a+1,b+1} - m_{1,b+1} - m_{1,a+1} + m_{1,1}.$$

Finally, for $1 \le a \le n-1$,

$$T(m_{a,a}) = m_{a+1,a+1} - 2m_{1,a+1} + m_{1,1}.$$

Writing these images as coordinate columns with respect to the ordered basis $\mathcal{B}_n$ gives precisely the blocks C_j, $R_i^{(j)}$, and $p^{(j)}$ described above. Hence the matrix of T in this basis is

$$A_n$$

as claimed. □

Example 2.4.2. For $n = 4$, the preceding block description gives

$$A_4 = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -1 & -2 & -1 & -1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -2 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -2 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}.$$

For $n = 5$, we obtain

$$A_5 = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -1 & -2 & -1 & -1 & 0 & -1 & -1 & 0 & 0 & -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -2 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -2 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 & -2 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Lemma 2.4.3. *For every $n \geq 2$, let*

$$B_n = \begin{pmatrix} 1 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & -1 & -1 & \cdots & -1 \end{pmatrix} \in M_n(\mathbb{C}).$$

Then B_n is similar to the permutation matrix. Hence,

$$\chi_{B_n}(t) = t^n - 1.$$

Proof. Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{C}^n$. By the basic rule of matrix multiplication, we have

$$B_n e_1 = e_1, \qquad B_n e_i = e_{i-1} - e_n \quad (2 \leq i \leq n).$$

Now let $S \colon \mathbb{C}^n \to \mathbb{C}^n$ be the cyclic permutation operator defined by

$$Se_1 = e_n, \qquad Se_i = e_{i-1} \quad (2 \leq i \leq n).$$

Thus S is represented in the standard basis by the permutation matrix of the cycle

$$e_1 \longmapsto e_n \longmapsto e_{n-1} \longmapsto \cdots \longmapsto e_1.$$

For $1 \leq i \leq n$, define

$$x_i = e_i + e_{i+1} + \cdots + e_n.$$

The vectors $x_1, \ldots, x_n$ form a basis of $\mathbb{C}^n$, since

$$e_i = x_i - x_{i+1} \quad (1 \leq i < n), \qquad e_n = x_n.$$

Moreover,

$$Sx_1 = Se_1 + Se_2 + \cdots + Se_n = e_n + e_1 + \cdots + e_{n-1} = x_1.$$

For $2 \leq i \leq n$, we have

$$\begin{aligned} Sx_i &= S(e_i + e_{i+1} + \cdots + e_n) \\ &= e_{i-1} + e_i + \cdots + e_{n-1} \\ &= x_{i-1} - x_n. \end{aligned}$$

Therefore, relative to the ordered basis $(x_1, \ldots, x_n)$, the columns of the matrix of S are

$$x_1, \quad x_1 - x_n, \quad x_2 - x_n, \quad \ldots, \quad x_{n-1} - x_n.$$

Hence

$$[S]_{(x_1,\ldots,x_n)} = B_n.$$

It follows that B_n is similar to the n-cycle permutation matrix $[S]_{(e_1,\ldots,e_n)}$.

Finally, e_1 is a cyclic vector for S: the vectors

$$e_1,\ Se_1,\ S^2e_1, \ldots, S^{n-1}e_1$$

are precisely the standard basis vectors in a different order. Therefore S has a cyclic vector, so $(t^n - 1)|\chi_{S_n}$. On the other hand, $S^n = I$, so the minimal polynomial divides $t^n - 1$. It follows that the minimal polynomial is $t^n - 1$. Since this polynomial has degree $n = \dim \mathbb{C}^n$, it is also the characteristic polynomial of S. Similar matrices have the same characteristic polynomial, so

$$\chi_{B_n}(t) = \chi_S(t) = t^n - 1.$$

□

Regard B_n as the linear transformation

$$B_n : \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

whose matrix in the standard basis is the matrix defined in Lemma 2.4.3. Recall that

$$\mathrm{Sym}^2(\mathbb{C}^n) = \frac{\mathbb{C}^n \otimes \mathbb{C}^n}{H},$$

where H is the subspace generated by

$$v \otimes w - w \otimes v \qquad (v, w \in \mathbb{C}^n).$$

The map B_n induces a linear operator

$$\rho(B_n) : \mathrm{Sym}^2(\mathbb{C}^n) \longrightarrow \mathrm{Sym}^2(\mathbb{C}^n)$$

defined on symmetric products by

$$\rho(B_n)(v \odot w) = (B_n v) \odot (B_n w), \qquad v, w \in \mathbb{C}^n,$$

and extended linearly.

Lemma 2.4.4. *Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{C}^n$, and set*

$$f_a = e_{n+1-a} \qquad (1 \leq a \leq n)$$

and

$$\begin{aligned} \mathcal{S}_n = \big(&f_n \odot f_n;\ f_{n-1} \odot f_n, f_{n-1} \odot f_{n-1}; \\ &f_{n-2} \odot f_n, f_{n-2} \odot f_{n-1}, f_{n-2} \odot f_{n-2};\ \ldots; \\ &f_1 \odot f_n, f_1 \odot f_{n-1}, \ldots, f_1 \odot f_1\big). \end{aligned}$$

Then $\mathcal{S}_n$ is an ordered basis of $\mathrm{Sym}^2(\mathbb{C}^n)$, and the matrix of the shorthandlinear operator

$$\rho(B_n) : \mathrm{Sym}^2(\mathbb{C}^n) \longrightarrow \mathrm{Sym}^2(\mathbb{C}^n)$$

with respect to this basis is A_n. In other words,

$$[\rho(B_n)]_{\mathcal{S}_n} = A_n.$$

Proof. Since $f_a = e_{n+1-a}$, the definition of B_n gives

$$B_n f_n = f_n$$

and, for $1 \leq a \leq n-1$,

$$B_n f_a = f_{a+1} - f_1.$$

We now compute the action of $\rho(B_n)$ on the symmetric products. First,

$$\begin{aligned}\rho(B_n)(f_n \odot f_n) &= (B_n f_n) \odot (B_n f_n) \\ &= f_n \odot f_n.\end{aligned}$$

For $1 \leq a \leq n-1$,

$$\begin{aligned}\rho(B_n)(f_a \odot f_n) &= (B_n f_a) \odot (B_n f_n) \\ &= (f_{a+1} - f_1) \odot f_n \\ &= f_{a+1} \odot f_n - f_1 \odot f_n.\end{aligned}$$

For $1 \leq a < b \leq n-1$,

$$\begin{aligned}\rho(B_n)(f_a \odot f_b) &= (f_{a+1} - f_1) \odot (f_{b+1} - f_1) \\ &= f_{a+1} \odot f_{b+1} - f_1 \odot f_{b+1} - f_1 \odot f_{a+1} + f_1 \odot f_1.\end{aligned}$$

Finally, for $1 \leq a \leq n-1$,

$$\begin{aligned}\rho(B_n)(f_a \odot f_a) &= (f_{a+1} - f_1) \odot (f_{a+1} - f_1) \\ &= f_{a+1} \odot f_{a+1} - 2f_1 \odot f_{a+1} + f_1 \odot f_1.\end{aligned}$$

□

3. Main Results

3.1. **Statement of the main result.**

Theorem 3.1.1. *Whenever n is an odd number greater than 3, $RSK_{(2n-2,2),(2^n)}$ has the characteristic polynomial $(t^n-1)^{(n+1)/2}$.*

Proof. Since B_n is similar to an n - cycle permutation matrix, it has eigenvalues $\xi, \xi^2, ..., 1$ where $\xi = e^{2\pi i/n}$. Now denote the corresponding eigenvectors by $v_1, v_2, .., v_n$. Hence in $Sym^2(V)$, we can build an ordered basis $\{v_i \odot v_j | 1 \leq i, j \leq n\}$, with eigenvalues ξ^{i+j} for any $1 \leq i, j \leq n$. Now we should count for a given integer $k \in \mathbb{Z}/n\mathbb{Z}$ the number of the solutions of $i + j \equiv k(mod\ n)$ for $1 \leq i, j \leq n$. Since n is an odd number, obviously the answer is $(n+1)/2$. Therefore each n-th root of unity appears as an eigenvalue of A_n with multiplicity $(n+1)/2$. Hence

$$\chi_{A_n}(t) = \prod_{k=0}^{n-1}(t - \zeta^k)^{(n+1)/2} = \left(\prod_{k=0}^{n-1}(t - \zeta^k)\right)^{(n+1)/2}.$$

Since

$$\prod_{k=0}^{n-1}(t - \zeta^k) = t^n - 1,$$

we obtain

$$\chi_{A_n}(t) = (t^n - 1)^{(n+1)/2}.$$

This proves the theorem. □

4. A Polynomial Family and Its Irreducibility

Since it is already known fixing the dimension of size of a matrix and the sum of all entires, the characteristic polynomial RSK matrix $RSK_{m,n,d}$ is not solvable by radicals. It is then natural to ask about the Galois group of the chracteristic polynomial of $RSK_{\sigma,\pi}$ fixing a pair of compositions σ, π.

4.1. **A computation Result.** *For $RSK(1,a,a),(a,a,1)$, considering the charatererstic polynomial of the matrix form of $RSK(1,a,a),(a,a,1)$ χ_a, it is checked that for $100 \geq a \geq 15$, $F_a(t) = \frac{t(t+2)(t^2+1)(t^2-1)^a+(-1)^a\left(t^4+2t^3-t+1-t^{2a+2}(t+2)\right)}{t^2-1} \mid \chi_a$ .Whats' more, $Gal(E_a/\mathbb{Q}) \cong S_{2a+2}$ where E_a is the splitting field of F_a over $\mathbb{Q}$.* One can compute the result using the following sage code:

4.2. **Irreducibility.**

Proposition 4.2.1. *$F_a(t)$ is irreducible over $\mathbb{Q}$ when $a \geq 100$.*

Proof. We denote $n = a-1$, $w = t^2$, $R = |w| = 1+h$, $d = |w-1|$, $e = |w+1|$. Here we fix t on a point in circle $|t+2| = 1$ in the complex space $\mathbb{C}$. We write $t = x+iy$, then we have

$$1 = |t+2|^2 = (x+2)^2 + y^2 = h+1+4x+4$$

. Then,

$$d^2 = |t-1|^2|t+1|^2 = (x^2+y^2-2x+1)(x^2+y^2+2x+1) = (h+1+2(1+h/4))(h+1-2(1+h/4)) = 2h+\frac{3h^2}{4}.$$

The real part of w, denoted by $\mathbf{Re}(w)$ and the imagine part of w, denoted by $\mathbf{Im}(w)$ can be expressed as follows

$$\mathbf{Re}(w) = x^2 - y^2 = 2x^2 - (x^2+y^2) = 2(-1-\frac{x}{4})^2 - (h+1) = 1+\frac{h^2}{8}$$

$$|\mathbf{Im}(w)|^2 = (h+1)^2 - |\mathbf{Re}(w)|^2 = d^2 - \frac{h^4}{64}$$

. We then denote

$$L_a(t) = (-1)^{a+1}(t+2)\sum_{k=2}^{a} t^{2k}.$$

$$E_a(t) = t(t+2)(t^2+1)(t^2-1)^{a-1} + (-1)^{a+1}(1-t+t^2+t^3).$$

It is easy to check $L_a(t) + E_a(t) = F_a(t)$. We shall estimate the module of $E_a(t)$ and $L_a(t)$.

$$|L_a(t)| = R^2\frac{|w^n-1|}{d}, |E_a(t)| \leq \sqrt{R}ed^n + e + \sqrt{R}d$$

. Case 1: We fix $\delta = \frac{1}{100}$. Then when $\delta \leq h \leq 8$, we take $q = \frac{2}{\sqrt{5}}, r = \frac{100}{101}$. On the compact interval $[\delta, 8]$,

$$\frac{d}{R} \leq q, \qquad \frac{1}{R} \leq r.$$

Therefore,

$$\begin{aligned}\frac{|E_a|}{|L_a|} &\leq \frac{\dfrac{ed}{R^{3/2}}\left(\dfrac{d}{R}\right)^n + \dfrac{(e+\sqrt{R}\,d)d}{R^2}R^{-n}}{1-R^{-n}} \\ &\leq \frac{6q^n+5r^n}{1-r^n} \leq 1 \qquad (a \geq 100),\end{aligned}$$

where we have used

$$e \leq 2R, \qquad d^2 \leq R^2-1, d \leq R, \qquad \sqrt{R} \leq 3.$$

Case 2: $0 \le h \le \delta$. Then we have

$$d < \frac{1}{7}, \sqrt{R} \le \frac{21}{20}, e = |w+1| \le R+1 < \frac{41}{40}$$

. Hence $|E_a(t)| < \frac{9}{4}$. We write $w = Re^{i\varphi}, \theta = |\varphi|$. Since $h \le \delta$, we have $\mathbf{Im}(w) \ge \frac{d^2}{2}$. Here $|R| \le 2$, hence we have

$$\theta \ge sin\theta \ge \frac{|\mathbf{Im}(w)|}{R} \ge \frac{d}{2}$$

$$cos\theta = \frac{1+\frac{h^2}{8}}{R} \ge \frac{1}{R}$$

.

$$\frac{h}{d} \ge \frac{R-1}{\sqrt{R^2-1}} \ge \sqrt{\frac{R-1}{R+1}} \ge \sqrt{\frac{1-\frac{1}{R}}{1+\frac{1}{R}}} \ge \sqrt{\frac{1-cos\theta}{1+cos\theta}} = tan\frac{\theta}{2} \ge \frac{\theta}{2}$$

Subcase 1: $n\theta \ge 5$**.**

Since $|w| = R > 1$, the reverse triangle inequality gives

$$|w^n - 1| \ge |w|^n - 1 = R^n - 1.$$

Moreover, since $h = R - 1$, or equivalently $R = 1 + h$, Bernoulli's inequality gives

$$R^n = (1+h)^n \ge 1 + nh = 1 + n(R-1).$$

Thus

$$R^n - 1 \ge n(R-1) = nh.$$

Using $R^2 \ge 1$ and the previous estimate $h/d \ge \theta/2$, we obtain

$$\begin{aligned}
|L_a(t)| &= R^2 \frac{|w^n - 1|}{d} \\
&\ge R^2 \frac{R^n - 1}{d} \\
&\ge \frac{R^n - 1}{d} \\
&\ge \frac{n(R-1)}{d} \\
&= \frac{nh}{d} \\
&\ge \frac{n\theta}{2} \\
&\ge \frac{5}{2}.
\end{aligned}$$

The last inequality follows from the assumption $n\theta \ge 5$.

Subcase 2: $n\theta \le 5$**.**

Write $w = Re^{i\varphi}$ and $\theta = |\varphi|$. Then

$$\begin{aligned}
|w^n - 1|^2 &= (R^n \cos(n\varphi) - 1)^2 + R^{2n} \sin^2(n\varphi) \\
&= R^{2n} - 2R^n \cos(n\varphi) + 1 \\
&= (R^n - 1)^2 + 2R^n(1 - \cos(n\varphi)) \\
&= (R^n - 1)^2 + 4R^n \sin^2 \frac{n\varphi}{2} \\
&= (R^n - 1)^2 + 4R^n \sin^2 \frac{n\theta}{2} \\
&\ge 4 \sin^2 \frac{n\theta}{2},
\end{aligned}$$

At the same time, it is known that

$$\frac{sin(n\theta/2)}{n\theta} \geq \frac{sin(5/2)}{5} \rightarrow 2sin(n\theta/2) > \frac{2sin\frac{5}{2}}{5}n\theta$$

. Hence

$$|w^n - 1| \geq \theta n/5$$
$$|L_a(t)| \geq \frac{\theta n/5}{d} \geq n/10 \geq 3.$$

In each case, we have proven that $|L_a(t)| > |E_a(t)|$ on $|t+2| = 1$ Denote the circle $|t+2| = 1$ by D. Now$F_a(t)$and$L_a(t)$ have the same number of zeros in D,hence obviously Fa(t) have one and only one zero in D. Suppose $F_a(t) = G_a(t)H_a(t)$,notice that $F_a(2) = (-1)^a$. Hence $|G_a(-2)| = 1;|H_a(-2)| = 1;G_a(t) \in \mathbb{Z}(t)$, $H_a(t) \in \mathbb{Z}(t)$ Since either $G_a(t)$ or $H_a(t)$ has the zero in D. Let us say that $G_a(t)$ has the zero. Then $|H(-2)| = \prod_{H_\alpha=0} |\alpha + 2| > 1$, Which leads to a contradiction. □

References

[But94] Lynne M. Butler. *Subgroup Lattices and Symmetric Functions*. American Mathematical Society, 1994.

[SS90] Bruce E. Sagan and Richard P. Stanley. Robinson–Schensted algorithms for skew tableaux. *Journal of Combinatorial Theory, Series A*, 55(2):161–193, 1990.

[Sta01] Richard P. Stanley. *Enumerative Combinatorics: Volume 2*. Cambridge University Press, first edition, 2001.

[SY26] Ada Stelzer and Alexander Yong. Rsk as a linear operator. *Selecta Mathematica*, 32(4):88, Aug 2026.

[WP23] Zhang Weilin and Yuan Pingzhi. A note on an irreducible class of polynomials over integers, 2023. arXiv:2303.03605.